

Base. Figure A.5 contains the complete segment-wise comparison.

The filter study evaluates every SOH model before post-processing, after stage 1, and after the final stage using an identical input order. Stage 1 yields MAE values of 1.677, 1.644, and 1.926 percentage points for Base, Pruned, and Quantized. The final values for this controlled sequence are 1.476, 1.695, and 1.028 percentage points. Accordingly, post-processing can alter both the numerical error and the ordering of the variants. Figure A.6 shows this interaction together with the response characteristics calculated in Equation 7.8. The final filter can reduce short-term variation substantially, but its EMA-only 90 % response time of 26.65 days at 1 Hz must be considered together with the reported accuracy.

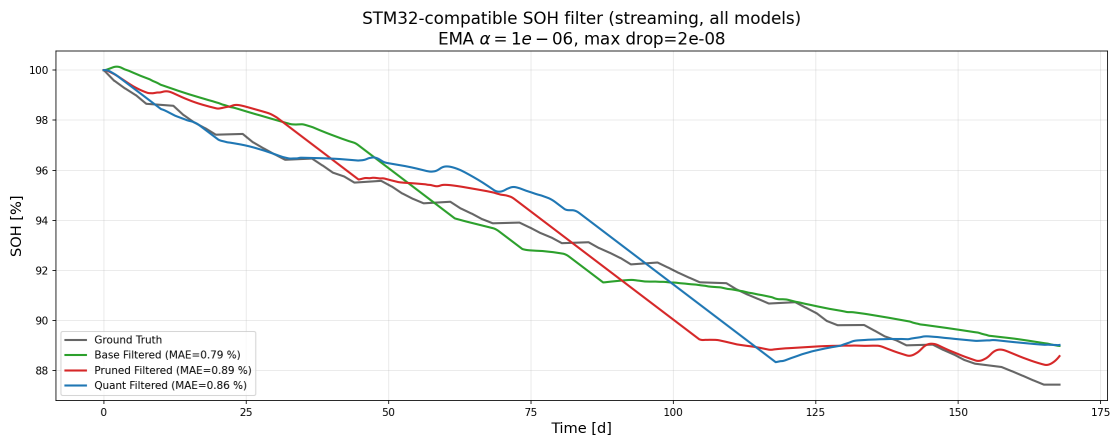


Figure 7.7: Streaming SOH dashboard with causal filtering for the three deployment variants. The upper panel shows the reference SOH together with the base, pruned, and quantized predictions after online smoothing, while the lower panel displays the corresponding prediction errors. The figure highlights the much slower degradation dynamics and the stronger role of causal post-processing in the SOH task.

7.9 Error Distribution and Local Transient Behaviour

Figures 7.8 and 7.9 analyse the statistical distribution of the prediction errors. For SOC, the distribution is narrow and approximately symmetric. The median absolute errors are 1,81 % for Pruned, 1,89 % for Base, and 2,03 % for Quantized. The pruning diagnostics in Figure A.4 verify that the 19 removed channels are the lowest-ranked channels under the defined L2 saliency criterion and compare the central recurrent-weight distributions before and after removal. These observations establish the implemented selection and the lower error of this particular Pruned